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CLOSED CELL SPRAY FOAM

Enverge OnePass HFO

Medium-Density Closed Cell Spray Polyurethane Foam · HFO blowing agent

Manufactured by **Holcim Solutions and Products US, LLC** · Manufactured in Waukesha, Wisconsin · Country of origin: **United States**

Enverge OnePass HFO is a spray-applied, two-component closed cell polyurethane foam (ASTM C1029 Type II) used to insulate and seal walls, attics, ceilings, crawl spaces, ducts and interior assemblies. It delivers the highest R-value per inch of any common insulation, stiffens the structure, and acts as an air barrier and Class II vapor retarder in a single pass.

R-7.2	>90%	2.1 lb/ft³	GWP 1
per inch, aged · R-28 at 4"	closed cell content	nominal density	HFO blowing agent
ASTM C518	ASTM D6226	ASTM D1622	Low global warming potential

WHY WE SPECIFY IT

1. **Highest R-value per inch**
   R-7.2 per inch aged, R-28 at 4 inches. In shallow cavities — rim joists, 2×4 walls, cathedral ceilings — closed cell reaches code R-values that open cell and batts cannot.

2. **Air barrier and vapor retarder in one layer**
   An air barrier material per ASTM E2178 at 1 inch or more (ABAA evaluated), and a Class II vapor retarder (≤ 1.0 perm) at 1-5/8 inches per ICC and ASHRAE.

3. **Tested to stay exposed in attics and crawl spaces**
   Passes ICC-ES AC377 Appendix X with no coating: up to 10 inches on walls and 12 inches on ceilings and slopes. Acceptance of an uncoated installation rests with the local building inspector; we confirm it with the inspector before the job and apply an approved coating where it is required.

4. **Built for the climate and for New England**
   HFO blowing agent with a global warming potential of 1, versus hundreds for older HFC foams. Rated for ambient service from -40°F to 200°F; stable through heat, cold and 100% humidity.

HEALTH & INDOOR AIR QUALITY

- **CDPH Standard 01350 compliant.** OnePass HFO is a low-VOC emitting material certified to the California Department of Public Health standard. This program applies strict certification criteria and builds in safety factors specifically to account for sensitive individuals such as children and the elderly, and certifies a product as acceptable for use in environments such as schools and healthcare facilities.
- **Referenced by CHPS and LEED.** The same 01350 certification is referenced by the Collaborative for High Performance Schools and the LEED Building Rating System as a low-emitting insulation credit.
- **No mold or fungal growth.** Tested to ASTM C1338 with no growth; closed cell foam absorbs almost no water (0.29% by volume).
- **Our standard practice with children in the home.** We recommend families with young children, elderly members, or anyone chemically sensitive plan to be out of the house during spraying and for a full 24–48 hours afterward while the foam finishes curing.

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TECHNICAL PROPERTIES

PROPERTY	VALUE	TEST METHOD
Apparent density	**2.1 lb/ft³ (±10%)**	ASTM D1622
R-Value (aged) at 75°F mean	**7.2 per inch · R-28 at 4"**	ASTM C518
Closed cell content	**> 90% by volume**	ASTM D6226
Compressive strength	**25 lb/in² nominal**	ASTM D1621
Tensile strength	**50 lb/in²**	ASTM D1623
Water absorption (96 h, 2" head)	**0.29% by volume**	ASTM D2842
Water vapor permeance at 1.6 in.	**≤ 1.0 perm**	ASTM E96
Air permeance at 75 Pa	**< 0.02 L/s·m²**	ASTM E2178
Fungi resistance	**No growth**	ASTM C1338
Dimensional stability at -20°F	**< 5% change**	ASTM D2126
Dimensional stability at +200°F	**< 10% change**	ASTM D2126
Dimensional stability, 158°F / 100% RH	**< 10% change**	ASTM D2126
Flame spread index (at 4 in.)	**Class 1 (A) — < 25**	ASTM E84
Smoke developed index (at 4 in.)	**Class 1 (A) — < 450**	ASTM E84
Global warming potential (blowing agent)	**1**	HFO
Service temperature range	**-40°F to 200°F**	ambient

CODE COMPLIANCE & CERTIFICATIONS

CSI master specification	07 21 19
Surface burning characteristics	ASTM E84 / UL 723 — Class 1 (A), per NFPA 101 and IBC Section 803
Ignition barrier	ICC-ES AC377 Appendix X — Pass with no coating up to 10" on walls and 12" on ceilings and slopes, subject to approval by the local building official (AHJ)
Thermal barrier	NFPA 286 — DC315 at 14 wet mils (up to 5.5" walls / 10.25" ceilings), No-Burn at 14 wet mils (6" / 9.5"), or Flame Control at 12 wet mils (6" / 10")
Air barrier · vapor retarder	ASTM E2178 at 1" or more, ASTM E2357 assembly, ABAA D-115-010 · Class II (≤ 1 perm) at 1-5/8"; Enverge recommends a minimum of 2"
Indoor air quality	CDPH Standard 01350 — low-emitting material; referenced by CHPS and LEED

ENVIRONMENTAL & CONTENT

VOC compliance
Low-emitting insulation, CA Section 01350

Recycled content
1.2% pre-consumer · 5.3% post-consumer

Rapidly renewable content
5.7%

MANUFACTURER DOCUMENTATION

Technical Data Sheet (TDS)	envergesprayfoam.com
Safety Data Sheet (SDS)	Parts A and B, on request
Architectural specification	CSI 07 21 19

Full TDS and SDS for every product on your project are available on request, at no charge, same day.

COATING & ASSEMBLY NOTE

OnePass HFO is tested (ICC-ES AC377 Appendix X) to remain exposed in attics and crawl spaces up to 10" on walls and 12" on ceilings and slopes with no coating. Whether an uncoated installation is accepted is decided by the local building official; Green Idea confirms this with the inspector before the job and applies an approved intumescent coating (DC315 / FireShell F10E) whenever it is required. Where the foam faces occupied space it must be separated by an approved 15-minute thermal barrier, such as 1/2" gypsum board, or by one of the NFPA 286 tested coatings listed above. Minimum installed thickness recommended by Enverge: 2".